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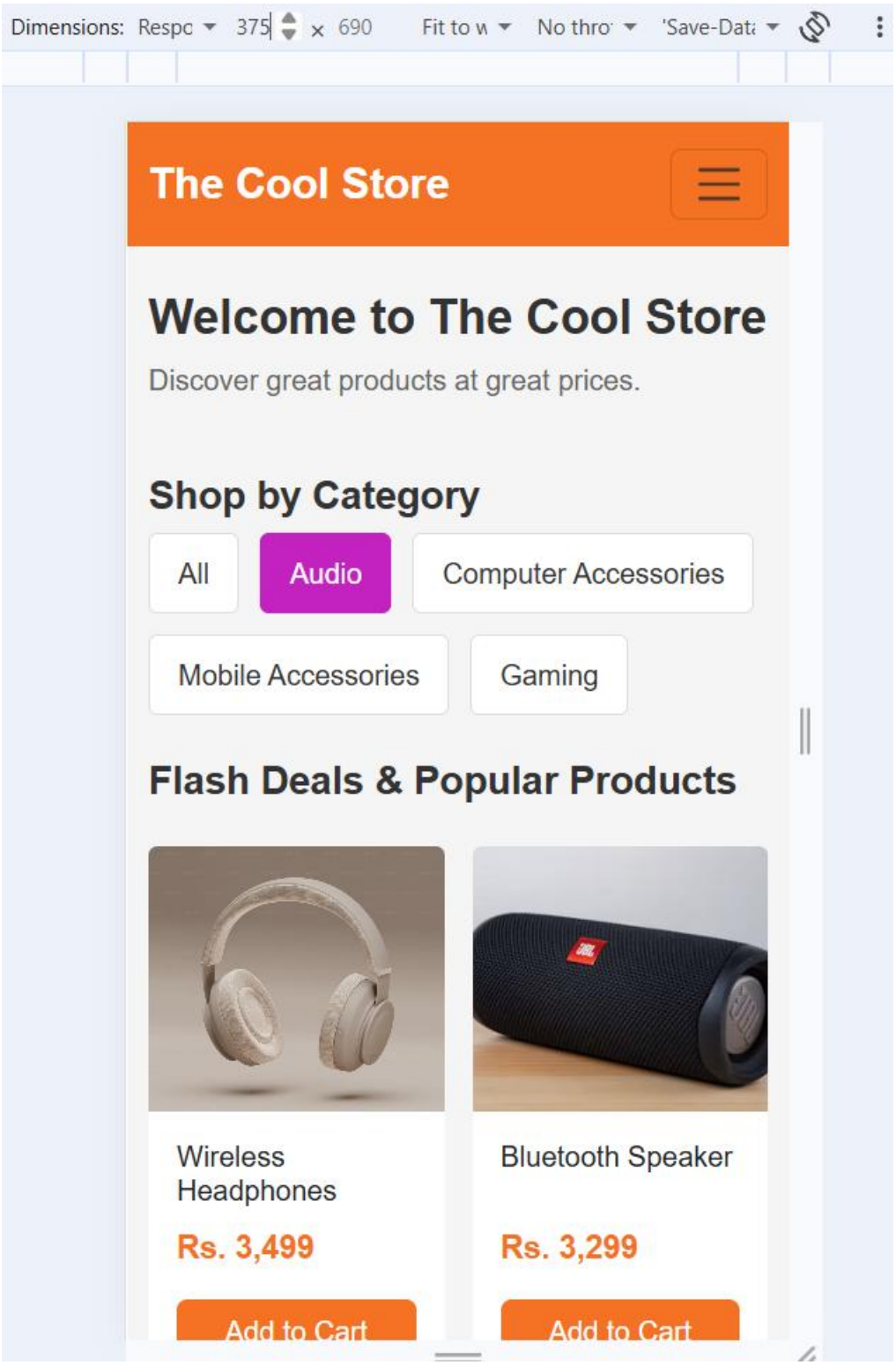
Web Technologies (CSC336) - Lab Mid Term, FA26 BCS A

N = 43, BP = 643px, H = 301 (accent hsl(301, 70%, 45%)), K = 4, sticky offset = 73px, Premium limit = Rs. 4,300

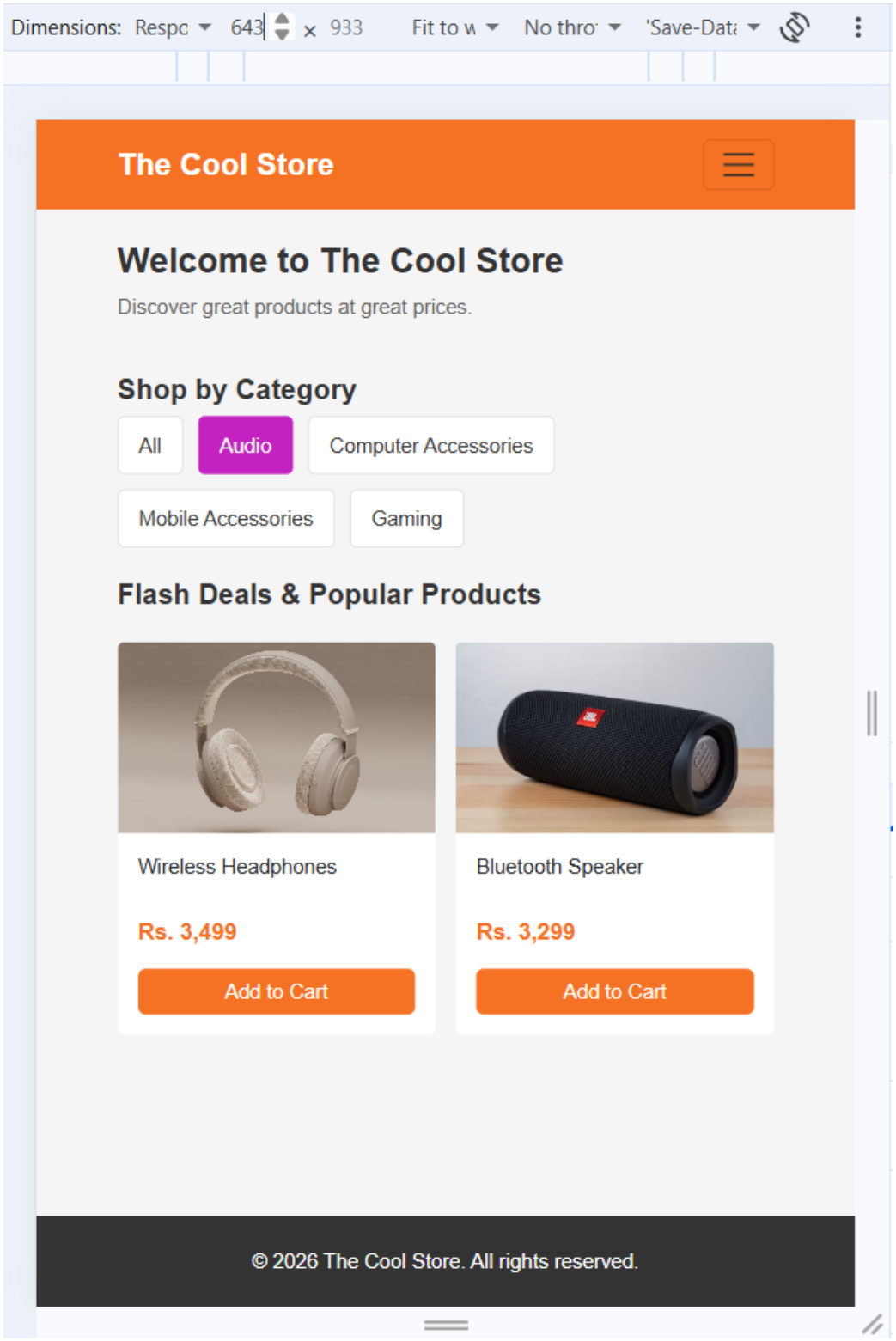
Task 1 - CSS-only category filter

Screenshots

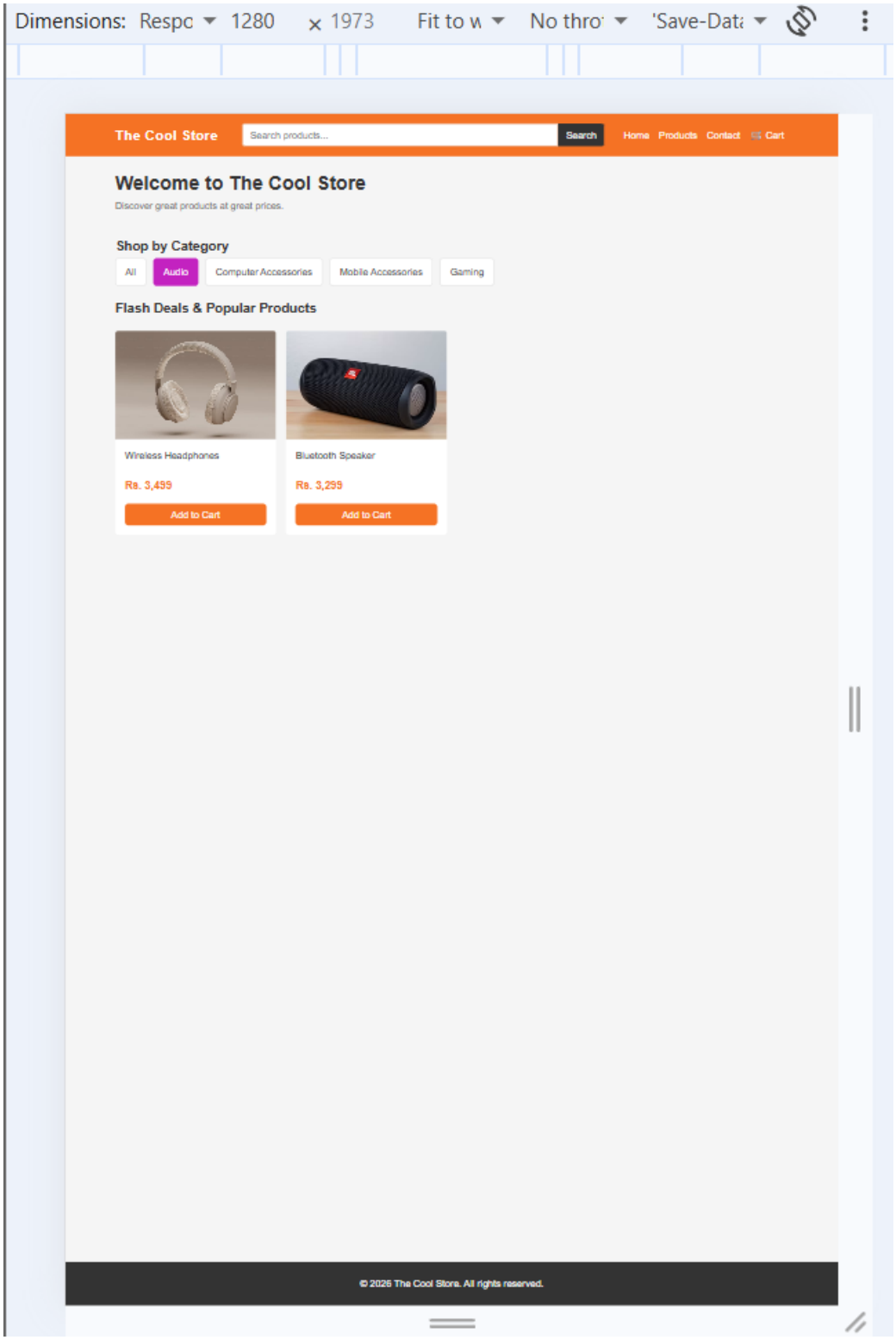
Task 1 at 375px (mobile)



Task 1 at 643px (BP)



Task 1 at 1280px (desktop)



Code

index.html - category radios and labels

```
<!-- Task 1: radios (hidden) + the existing category buttons turned into labels -->
      <input type="radio" name="category" id="cat-all" value="all" class="visually-
hidden" checked />
      <input type="radio" name="category" id="cat-audio" value="audio" class="visually-
hidden" />
```

```
      <input type="radio" name="category" id="cat-computer" value="computer"
class="visually-hidden" />
      <input type="radio" name="category" id="cat-mobile" value="mobile"
class="visually-hidden" />
      <input type="radio" name="category" id="cat-gaming" value="gaming"
class="visually-hidden" />

      <div class="category-list">
        <label for="cat-all">All</label>
        <label for="cat-audio">Audio</label>
        <label for="cat-computer">Computer Accessories</label>
        <label for="cat-mobile">Mobile Accessories</label>
        <label for="cat-gaming">Gaming</label>
      </div>
```

index.html - product cards (category added to each card)

```
<div class="col-6 col-md-4 col-lg-3 product-item" data-category="audio">
  ... (same change on all 8 cards: audio, computer, computer, mobile, mobile, computer,
gaming, audio)
```

style.css

```
/* ===== MIDTERM Task 1 (H = 301) ===== */
:root {
  --accent: hsl(301, 70%, 45%);
}

/* the category buttons are now labels, styled like the old links */
.category-list label {
  background-color: white;
  color: #333;
  padding: 10px 16px;
  border-radius: 5px;
  border: 1px solid #ddd;
  cursor: pointer;
}

.category-list label:hover {
  color: var(--accent);
  border-color: var(--accent);
}

/* selected filter uses the accent colour */
body:has(#cat-all:checked) label[for="cat-all"],
body:has(#cat-audio:checked) label[for="cat-audio"],
body:has(#cat-computer:checked) label[for="cat-computer"],
body:has(#cat-mobile:checked) label[for="cat-mobile"],
body:has(#cat-gaming:checked) label[for="cat-gaming"] {
  background-color: var(--accent);
  border-color: var(--accent);
  color: white;
}

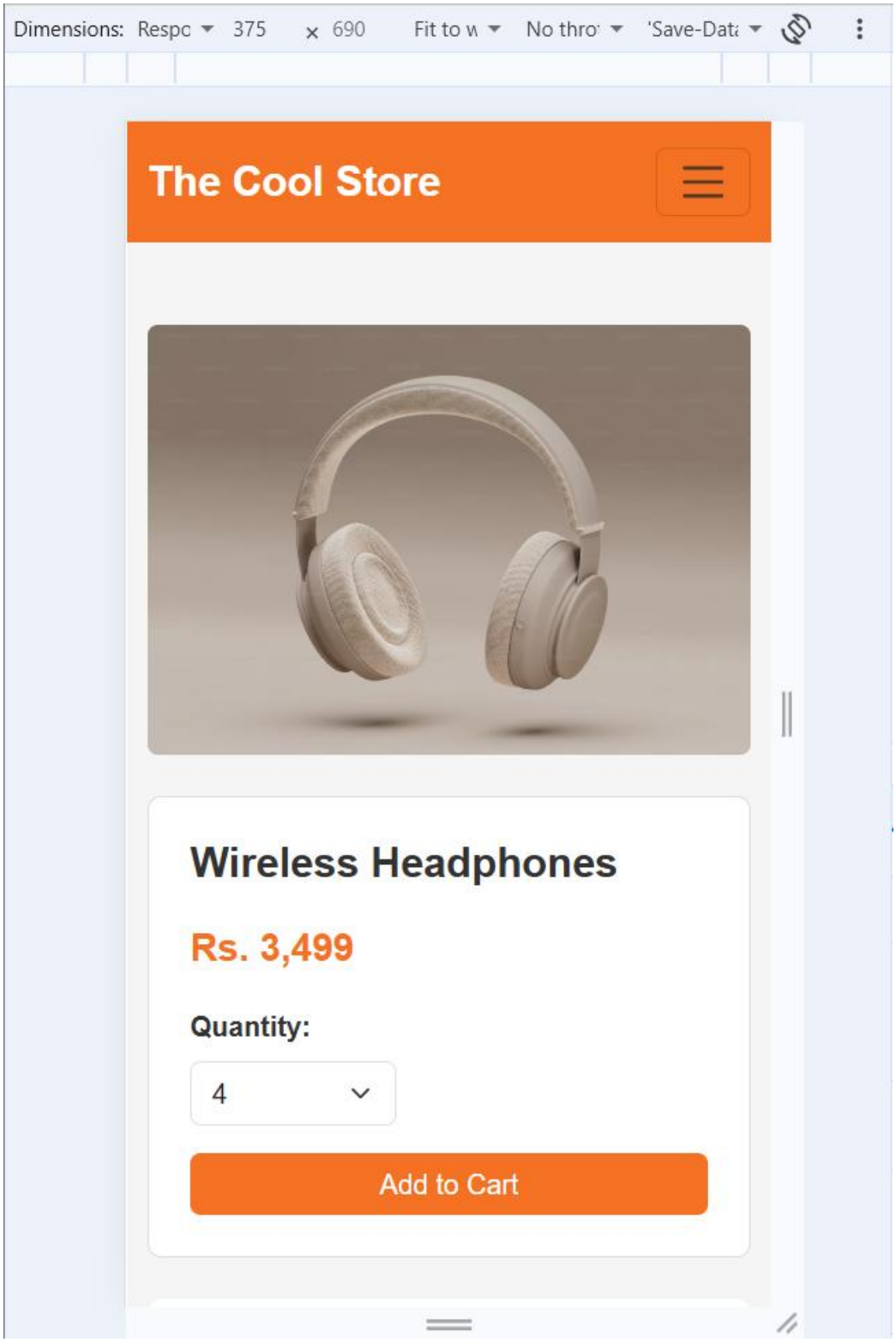
/* hide cards that don't match (display:none = no gaps in the grid) */
body:has(#cat-audio:checked) .product-item:not([data-category="audio"]),
body:has(#cat-computer:checked) .product-item:not([data-category="computer"]),
body:has(#cat-mobile:checked) .product-item:not([data-category="mobile"]),
body:has(#cat-gaming:checked) .product-item:not([data-category="gaming"]) {
  display: none;
}

/* small phones: same smaller buttons your old links had */
@media (max-width: 375px) {
  .category-list label {
    padding: 8px 10px;
    font-size: 14px;
  }
}
```

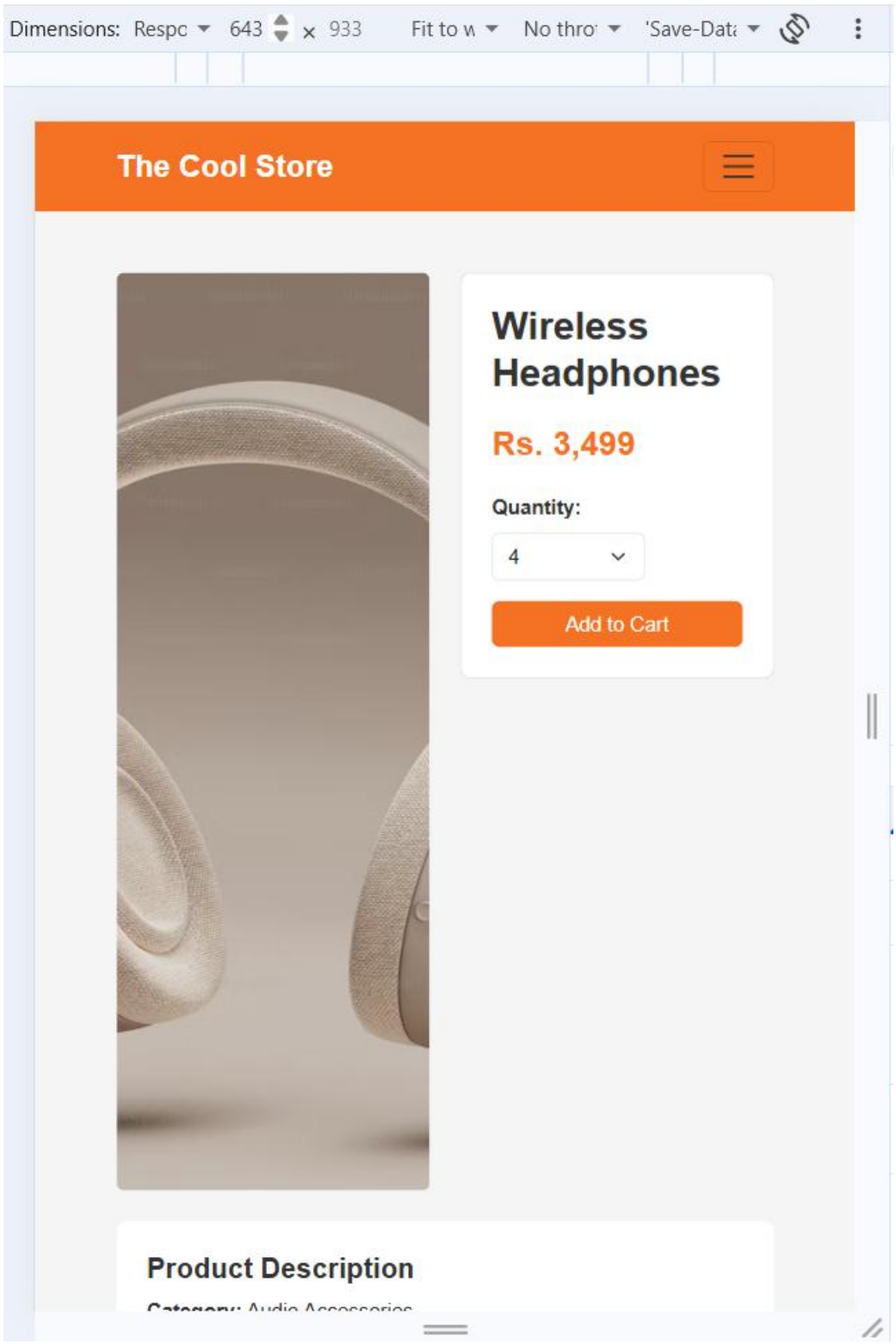
Task 2 - Product page grid with sticky details box

Screenshots

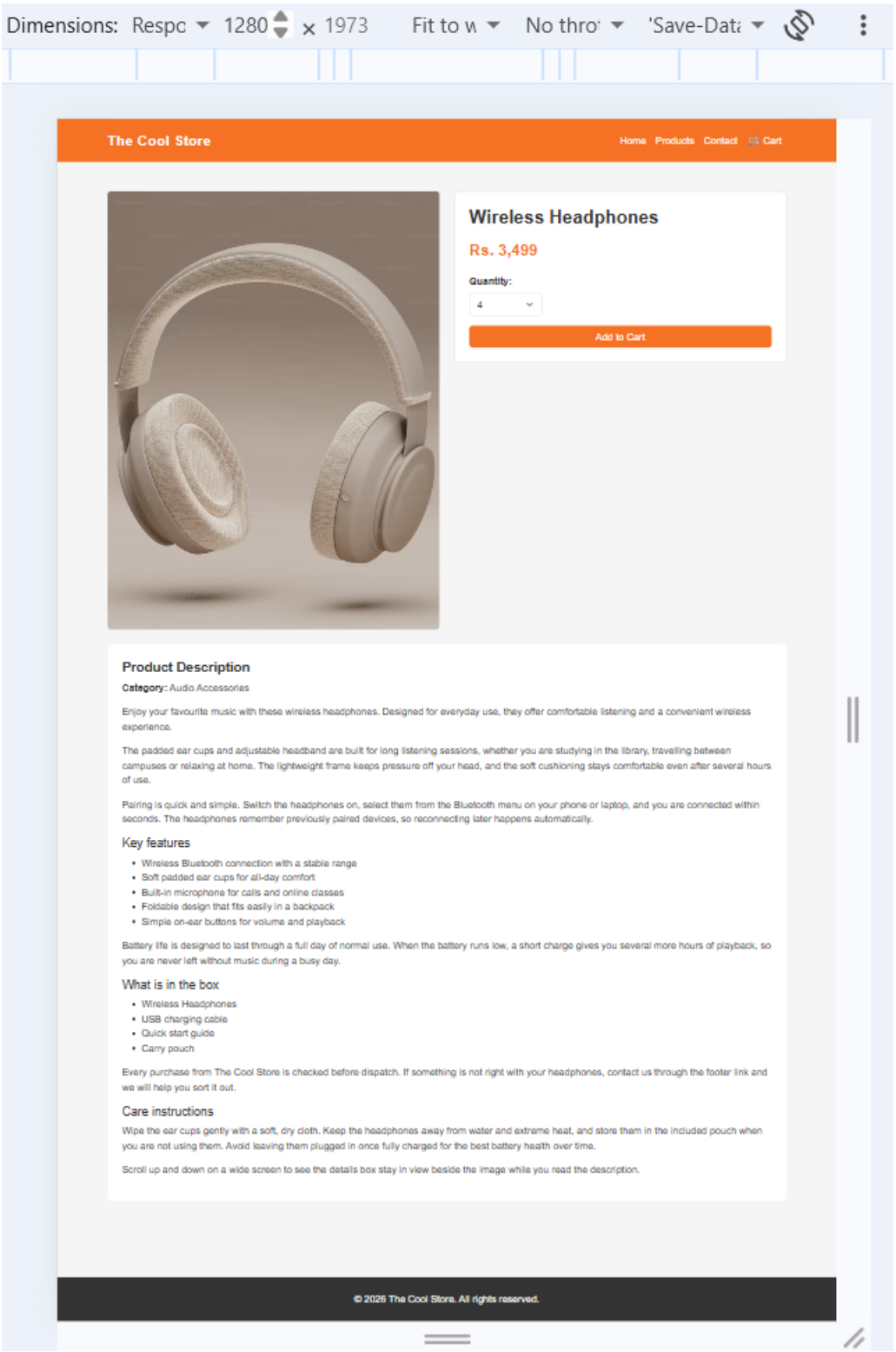
Task 2 at 375px (mobile)



Task 2 at 643px (BP)



Task 2 at 1280px (desktop)



Code

product.html - main section

```
<!-- Product Details (Task 2: CSS Grid with named areas) -->
    <main class="container my-5">
      <div class="product-layout">
        <!-- grid-area: image -->
        <div class="product-image">
          <img
```

```
            src="https://plus.unsplash.com/premium_photo-1679513691474-
73102089c117?q=80&w=1113&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=NHwxMjA3fDA%3D"
            alt="Wireless Headphones"
            class="img-fluid rounded product-detail-image"
          />
        </div>

        <!-- grid-area: details (sticky from BP up) -->
        <div class="product-details">
          <h1 class="product-detail-title">Wireless Headphones</h1>

          <p class="price product-detail-price">Rs. 3,499</p>

          <label for="quantity" class="form-label">
            <strong>Quantity:</strong>
          </label>

          <select id="quantity" class="form-select quantity-select">
            <option value="1">1</option>
            <option value="2">2</option>
            <option value="3">3</option>
            <option value="4">4</option>
            <option value="5">5</option>
          </select>

          <button type="button" class="btn add-btn mt-3">Add to Cart</button>
        </div>

        <!-- grid-area: description -->
        <section class="product-description">
          <h2 class="section-heading">Product Description</h2>
          <p><strong>Category:</strong> Audio Accessories</p>
          <p>
            Enjoy your favourite music with these wireless headphones. Designed
            for everyday use, they offer comfortable listening and a convenient
            wireless experience.
          </p>
          <p>
            The padded ear cups and adjustable headband are built for long
            listening sessions, whether you are studying in the library,
            travelling between campuses or relaxing at home. The lightweight
            frame keeps pressure off your head, and the soft cushioning stays
            comfortable even after several hours of use.
          </p>
          <p>
            Pairing is quick and simple. Switch the headphones on, select them
            from the Bluetooth menu on your phone or laptop, and you are
            connected within seconds. The headphones remember previously paired
            devices, so reconnecting later happens automatically.
          </p>
          <h3 class="h5">Key features</h3>
          <ul>
            <li>Wireless Bluetooth connection with a stable range</li>
            <li>Soft padded ear cups for all-day comfort</li>
            <li>Built-in microphone for calls and online classes</li>
            <li>Foldable design that fits easily in a backpack</li>
            <li>Simple on-ear buttons for volume and playback</li>
          </ul>
          <p>
            Battery life is designed to last through a full day of normal use.
            When the battery runs low, a short charge gives you several more
            hours of playback, so you are never left without music during a
            busy day.
          </p>
          <h3 class="h5">What is in the box</h3>
          <ul>
            <li>Wireless Headphones</li>
            <li>USB charging cable</li>
            <li>Quick start guide</li>
            <li>Carry pouch</li>
          </ul>
          <p>
            Every purchase from The Cool Store is checked before dispatch. If
```

```
            something is not right with your headphones, contact us through the
            footer link and we will help you sort it out.
          </p>
          <h3 class="h5">Care instructions</h3>
          <p>
            Wipe the ear cups gently with a soft, dry cloth. Keep the
            headphones away from water and extreme heat, and store them in the
            included pouch when you are not using them. Avoid leaving them
            plugged in once fully charged for the best battery health over
            time.
          </p>
          <p>
            Scroll up and down on a wide screen to see the details box stay in
            view beside the image while you read the description.
          </p>
        </section>
      </div>
    </main>
```

style.css

```
/* ===== MIDTERM Task 2 (BP = 643px, sticky top = 73px) ===== */
.product-layout {
  display: grid;
  gap: 24px;
  grid-template-columns: minmax(0, 1fr);
  grid-template-areas:
    "image"
    "details"
    "description";
}

.product-image {
  grid-area: image;
}

.product-details {
  grid-area: details;
  background-color: white;
  border: 1px solid #ddd;
  border-radius: 8px;
  padding: 24px;
}

.product-description {
  grid-area: description;
  background-color: white;
  border-radius: 8px;
  padding: 24px;
}

/* 643px = BP */
@media (min-width: 643px) {
  .product-layout {
    grid-template-columns: repeat(2, minmax(0, 1fr));
    grid-template-areas:
      "image details"
      "description description";
  }

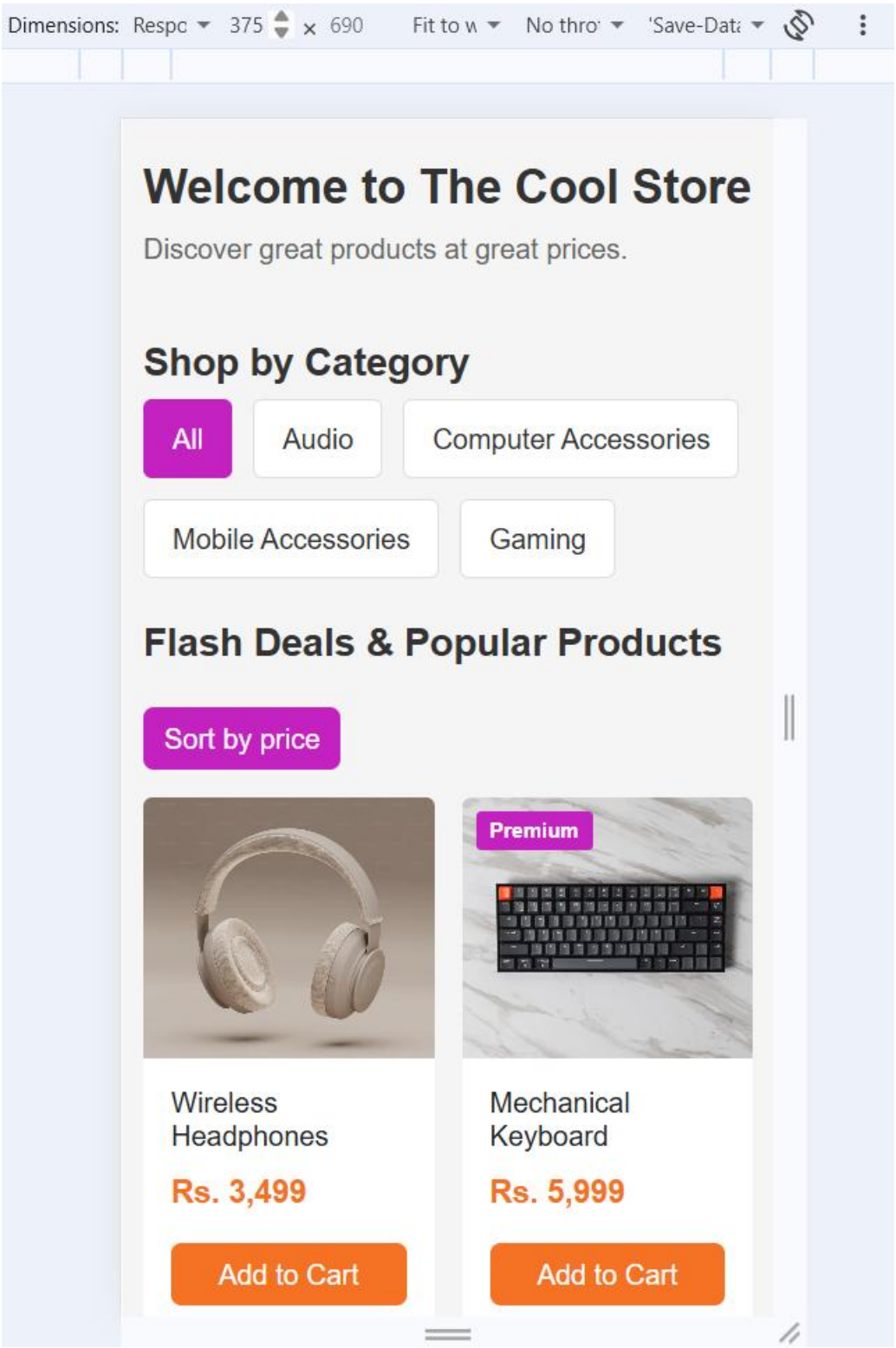
  .product-details {
    position: sticky;
    top: 73px; /* (43 mod 40) + 70 */
    align-self: start; /* without this the box stretches and cannot stick */
  }

  /* tall image row gives the sticky box room to travel */
  .product-image .product-detail-image {
    max-height: none;
    height: clamp(520px, 85vh, 720px);
  }
}
```

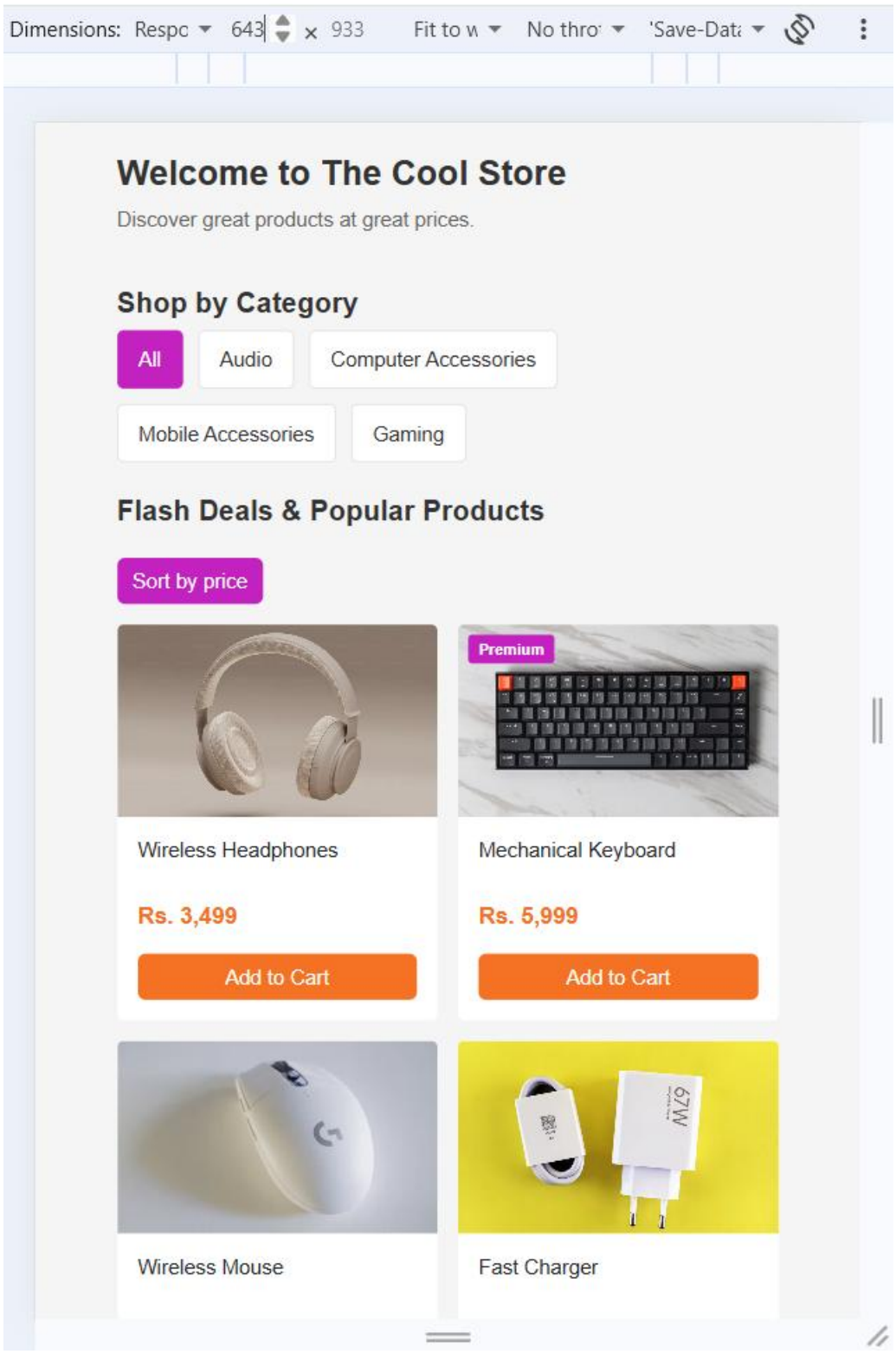
Task 3 - Live search, price sort and Premium label

Screenshots

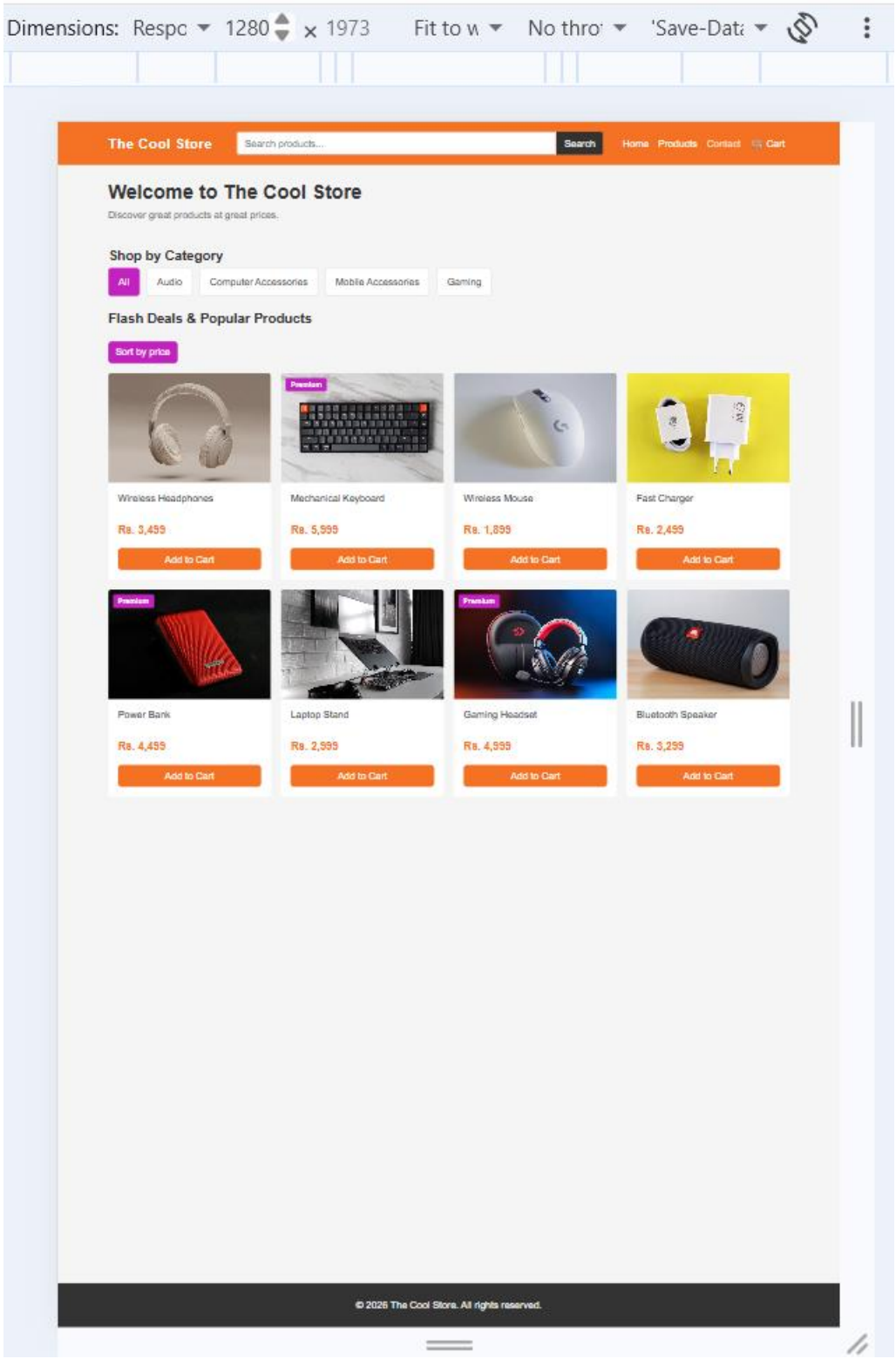
Task 3 at 375px (mobile)



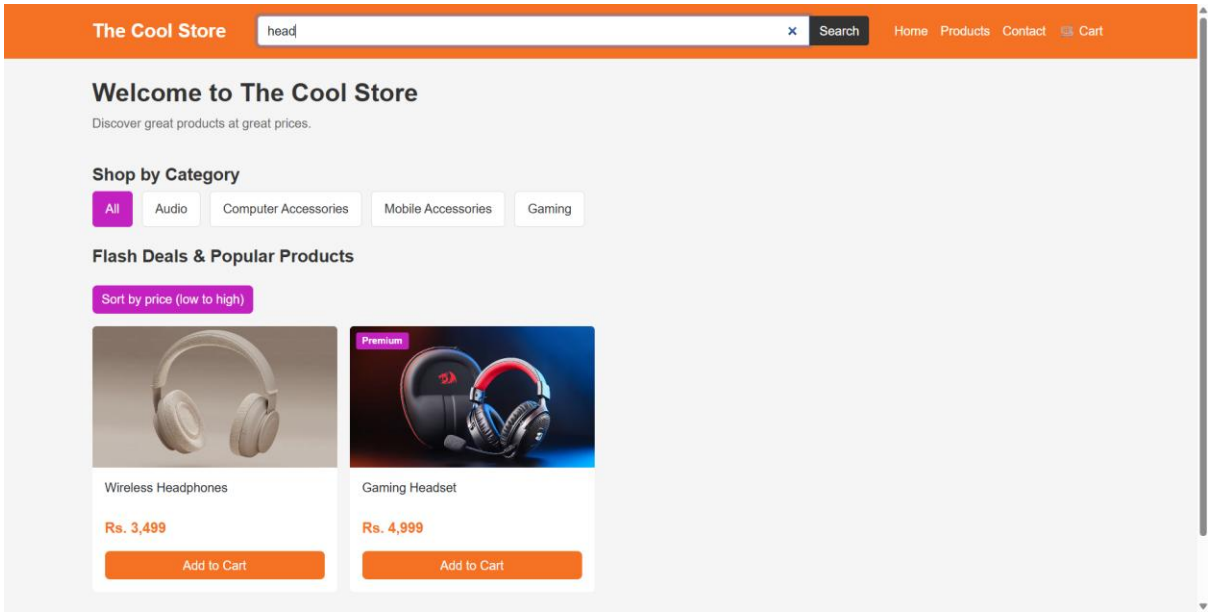
Task 3 at 643px (BP)



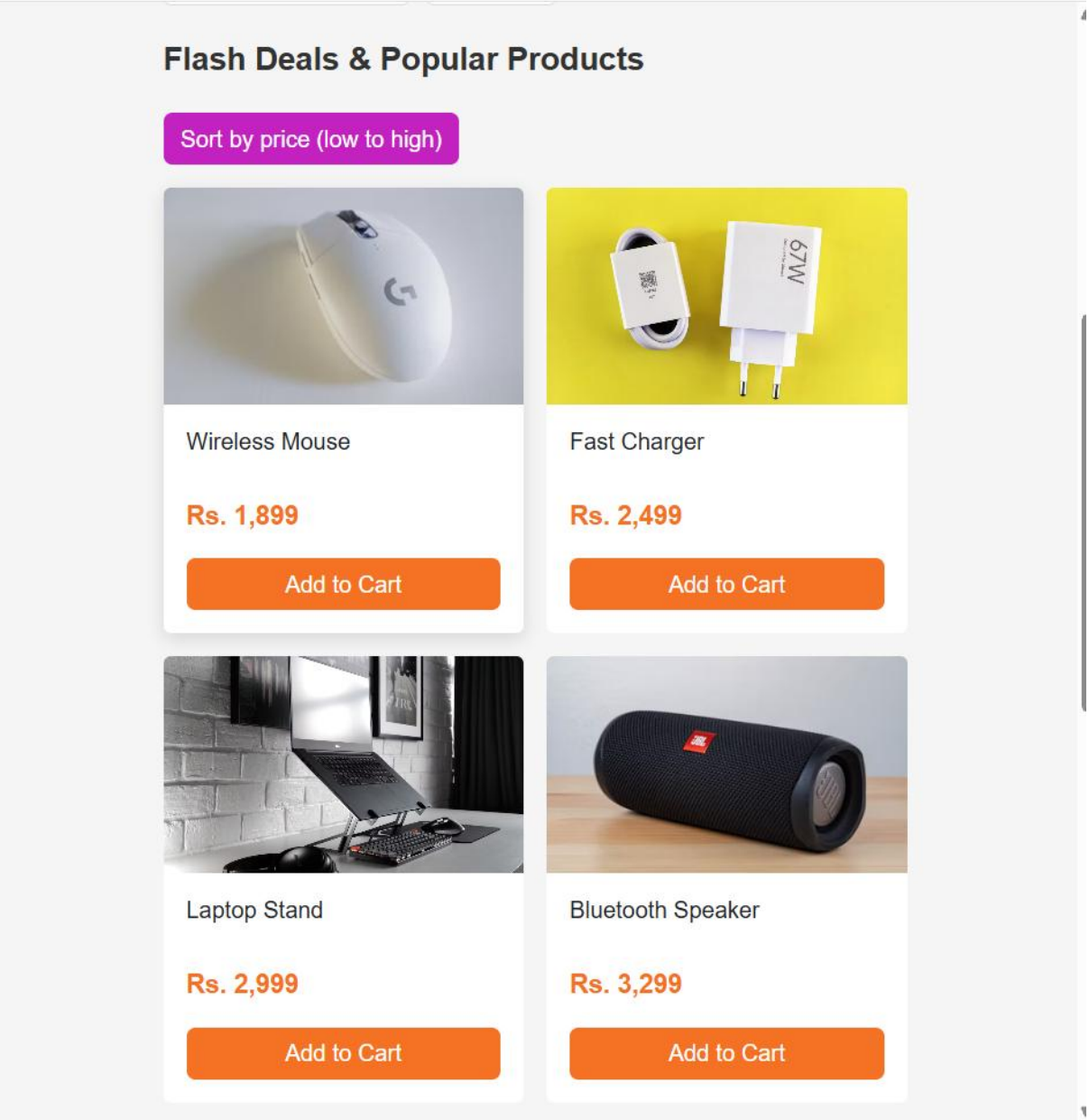
Task 3 at 1280px (desktop)



Task 3 after typing in the search box



Task 3 after clicking Sort by price



Code

index.html - navbar search form (existing search box given an id)

```
<form id="navSearchForm" class="d-flex search-form mx-lg-4 my-3 my-lg-0" role="search">
            <input
              id="productSearch"
              class="form-control"
              type="search"
              placeholder="Search products..."
              aria-label="Search products"
            />
            <button class="btn search-btn" type="submit">Search</button>
</form>
```

index.html - sort / reset buttons and the No products found message

```
<!-- Task 3: sort button (search = navbar search box) -->
      <div class="toolbar mb-3">
        <button type="button" id="sortBtn" class="btn sort-btn">Sort by price</button>
        <button type="button" id="resetBtn" class="btn sort-btn">Reset order</button>
```

```
      </div>

      <p id="noResults" class="no-results" hidden>No products found</p>
```

index.html - script link (before closing body tag)

```
<script src="script.js"></script>
```

script.js

```
// Lab Mid - Task 3 (N = 43)
const PREMIUM_LIMIT = 43 * 100; // N x 100 = 4300

const grid = document.querySelector(".product-grid");
const searchInput = document.getElementById("productSearch");
const sortBtn = document.getElementById("sortBtn");
const noResults = document.getElementById("noResults");
const radios = document.querySelectorAll('input[name="category"]');

// always read the cards from the page, so the order after sorting is respected
function getItems() {
  return Array.from(grid.querySelectorAll(".product-item"));
}

// "Rs. 3,499" -> 3499
function getPrice(item) {
  const text = item.querySelector(".price").textContent;
  return parseInt(text.replace(/[^\d]/g, ""), 10);
}

// remember the original card order (as written in the HTML)
const originalOrder = getItems();

// ---- Premium label (runs on page load) ----
getItems().forEach((item) => {
  if (getPrice(item) > PREMIUM_LIMIT) {
    const badge = document.createElement("span");
    badge.className = "premium-badge";
    badge.textContent = "Premium";
    item.querySelector(".product-card").appendChild(badge);
  }
});

// ---- Search (works together with the CSS category filter) ----
function applyFilters() {
  const query = searchInput.value.trim().toLowerCase();
  const category = document.querySelector('input[name="category"]:checked').value;
  let visibleCount = 0;

  getItems().forEach((item) => {
    const name = item.querySelector(".card-title").textContent.toLowerCase();
    const matchesName = name.includes(query);
    const matchesCategory = category === "all" || item.dataset.category === category;

    // the category filter hides cards with CSS; search hides them with this class.
    // A card shows only if neither hides it => it matches both.
    item.classList.toggle("search-hidden", !matchesName);

    if (matchesName && matchesCategory) {
      visibleCount++;
    }
  });

  noResults.hidden = visibleCount !== 0;
}

// the navbar search form must not reload the page
```

```
document.getElementById("navSearchForm").addEventListener("submit", (e) =>
e.preventDefault());

searchInput.addEventListener("input", applyFilters);
radios.forEach((radio) => radio.addEventListener("change", applyFilters));

// ---- Sort by price (toggles low->high, high->low) ----
let ascending = true;

sortBtn.addEventListener("click", () => {
  const sorted = getItems().sort((a, b) =>
    ascending ? getPrice(a) - getPrice(b) : getPrice(b) - getPrice(a)
  );

  // appendChild MOVES the existing elements, nothing is rebuilt
  sorted.forEach((item) => grid.appendChild(item));

  sortBtn.textContent = ascending ? "Sort by price (low to high)" : "Sort by price (high
to low)";
  ascending = !ascending;
});

// ---- Reset order (puts the cards back as they were on page load) ----
document.getElementById("resetBtn").addEventListener("click", () => {
  originalOrder.forEach((item) => grid.appendChild(item));
  ascending = true; // next sort click starts from low to high again
  sortBtn.textContent = "Sort by price";
});
```

style.css

```
/* ===== MIDTERM Task 3 (accent H = 301) ===== */
.sort-btn {
  background-color: var(--accent);
  color: white;
  border: none;
}

.sort-btn:hover {
  background-color: hsl(301, 70%, 35%);
  color: white;
}

/* cards hidden by the search (category filter hides with its own rules) */
.search-hidden {
  display: none !important;
}

.no-results {
  font-size: 18px;
  font-weight: bold;
  color: #666;
  text-align: center;
  padding: 40px 0;
}

.premium-badge {
  position: absolute;
  top: 8px;
  left: 8px;
  z-index: 1;
  background-color: var(--accent);
  color: white;
  font-size: 12px;
  font-weight: bold;
  padding: 2px 8px;
  border-radius: 3px;
}

.toolbar .sort-btn + .sort-btn {
```

```
  margin-left: 8px;
}
```